Yuchen Hou

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Summary

Master student at the National University of Singapore, focusing on Embodied AI and autonomous robotic decision-making.

Education

M.Sc. **National University of Singapore**, Electrical Engineering — Singapore, Aug 2025 – present

- Core member of the NUS CORE Lab (COntrol, Robotics & Artificial IntElligence), focusing on Embodied AI and autonomous robotic decision-making.
- Research on perception and planning for aerial robots based on Vision-Language Navigation (VLN), and end-to-end Reinforcement Learning in complex 3D environments.

B.Eng. **Nanjing University of Aeronautics and Astronautics**, Electronic and Communication Engineering — Nanjing, China, Sept 2021 – July 2025

- GPA: 88/100
- Key Courses: Advanced Mathematics (96), Probability Theory (97), Random Signal Analysis (97), Microwave Technology and Antenna (97), Fundamentals of Communication (95), Complex Function (99).

Awards

Second-class Scholarship: Nanjing University of Aeronautics and Astronautics (2024)

Third-class Scholarship: Nanjing University of Aeronautics and Astronautics (2021 – 2023)

Skills

Programming & Systems: Python (ROS 2, PyTorch, OpenCV), Linux (Ubuntu), Shell scripting

AI Frameworks & Simulation: PyTorch (DRL policies), Habitat, AI2-THOR (Complex 3D environments)

Robotics Development: ROS2 (rclpy, Nav2), Sensor Fusion, Lidar/RGB-D Integration, SLAM (Cartographer)

Research Capabilities: Sim-to-Real Transfer, Embodied AI Architectures, RL Algorithms, Academic Paper Writing, Tracking CVPR/IROS/ICRA

Languages

English: Proficient — IELTS 7.0 (R 8.5, W 7.5, L 6.5, S 6.0)

Mandarin: Native speaker